

Supervised ML

Regression

Classification

Two Categories
Binary classification

price
Salary
Temp.
etc

Continuous
value

Linear
Regression

Logistic
Regression

Category

class
1

class
0

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$$

$$f(\hat{y}) = p(y=1)$$

probability

price Target
 $> 1M$ (Exp.)
 $\leq 1M$ (Not exp.)

Example

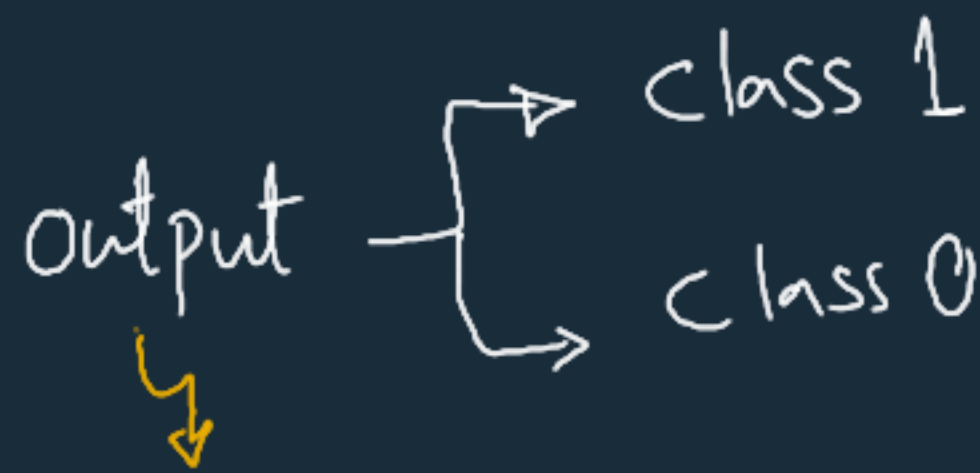
Target
 Expensive (1)
 Not expensive (0)

pred. = $p(\text{Expensive})$
 0 $\rightarrow$ 1

Threshold
0.5

$P(\text{Exp}) \geq 0.5$
pred = Exp? (1)
 $P(\text{Exp}) < 0.5$
pred = Not exp? (0)

Logistic Regression



click	positive	Spam
Not	Negative	Not spam

$p(\text{class 1})$

$0 \rightarrow 1$

Cont.



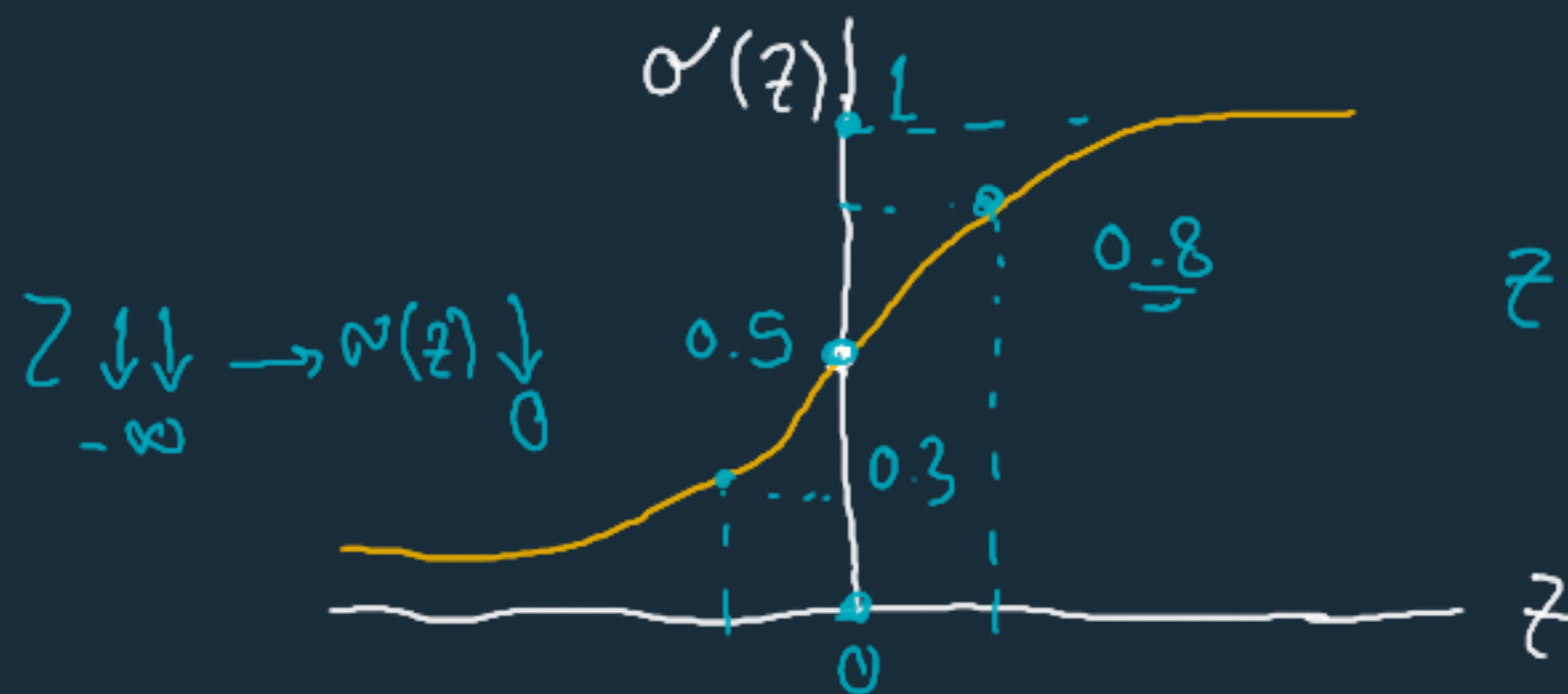
$0 \rightarrow 1$

$\sigma(z)$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots$$

Cont

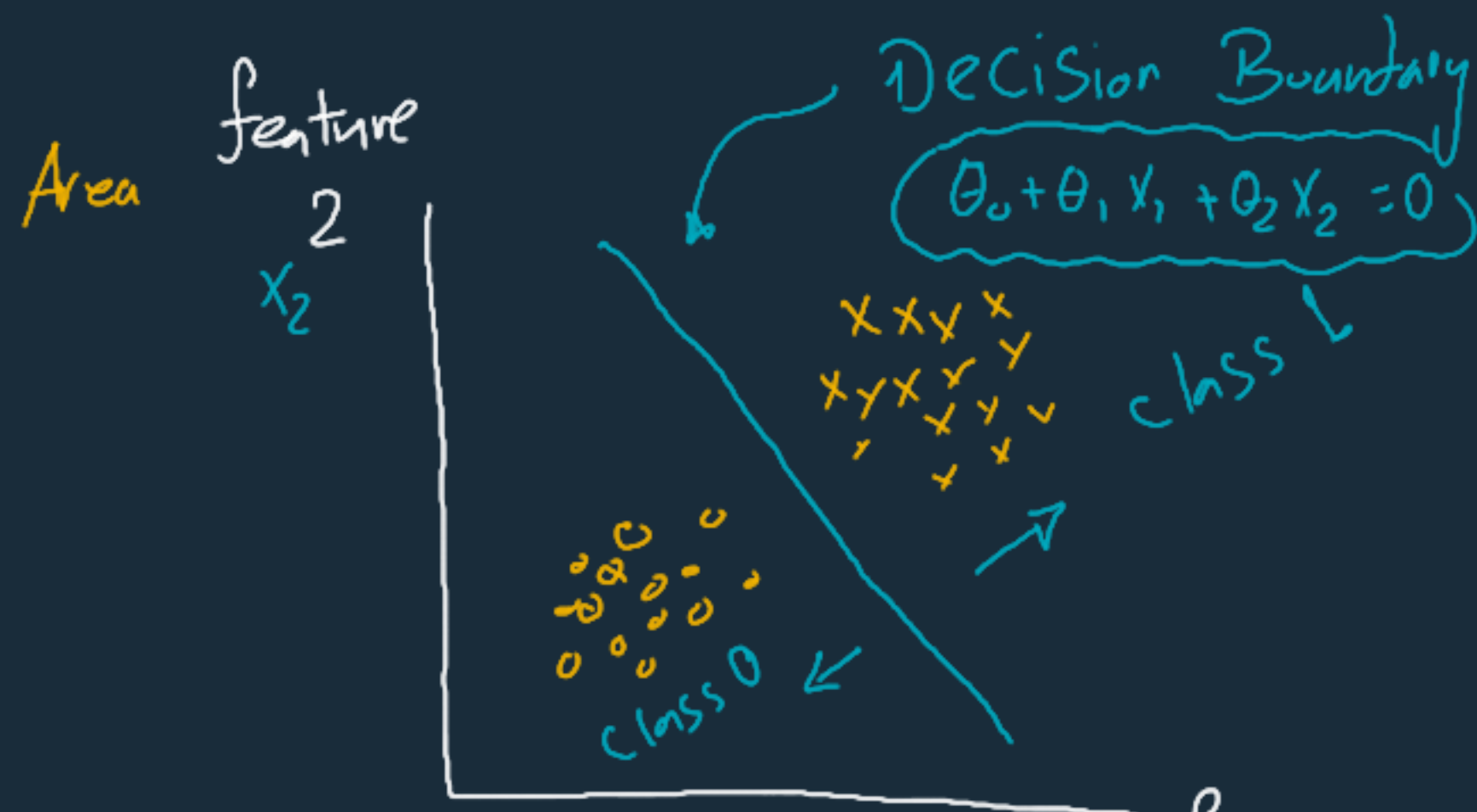


$z \downarrow \downarrow \downarrow -\infty \rightarrow \sigma(z) \downarrow 0$

$z \uparrow \uparrow \rightarrow \sigma(z) \uparrow 1$

$$z \geq 0 \rightarrow \text{class 1}$$

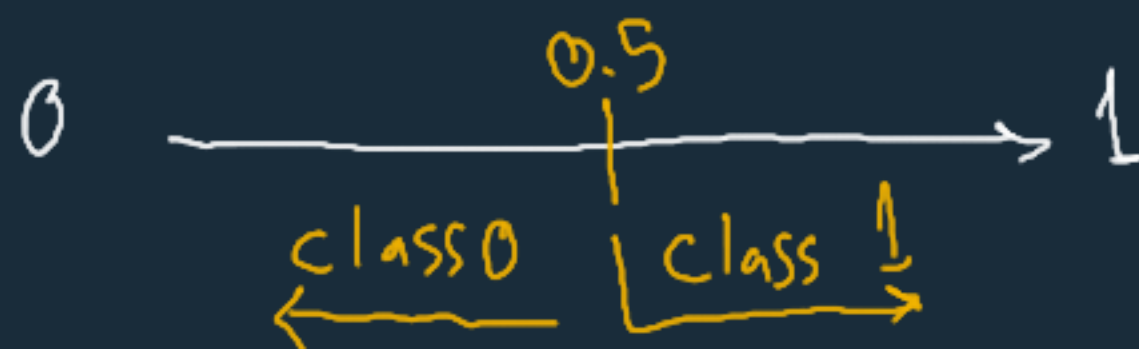
$$z < 0 \rightarrow \text{class 0}$$



Logistic Regression

min. Cost fn

$$p(\text{class 1}) = \sigma(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n)$$



Cost function

Log. Reg

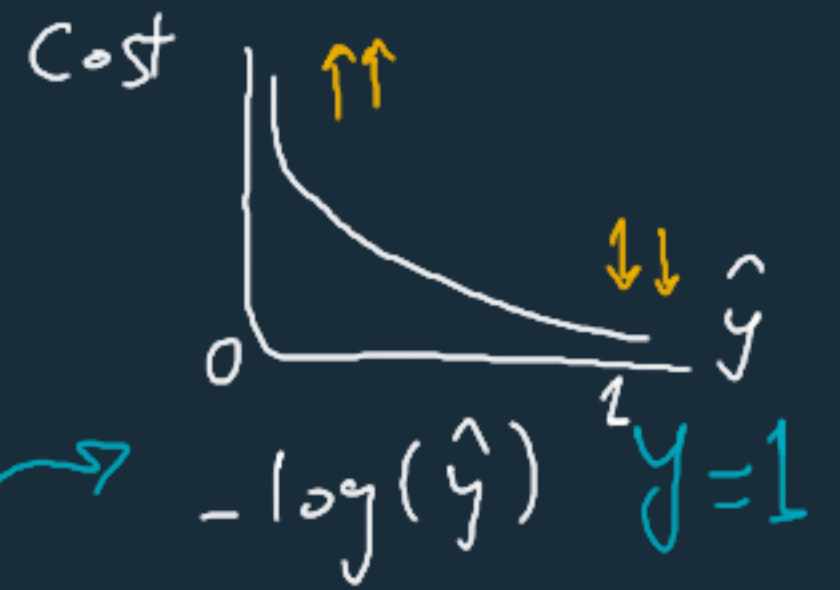
Logistic Regression Training

	Actual	pred	Cost
Cost $\rightarrow$	{ 1	1	$\downarrow$
		0	$\uparrow$
	{ 0	0	$\downarrow$
		1	$\uparrow$

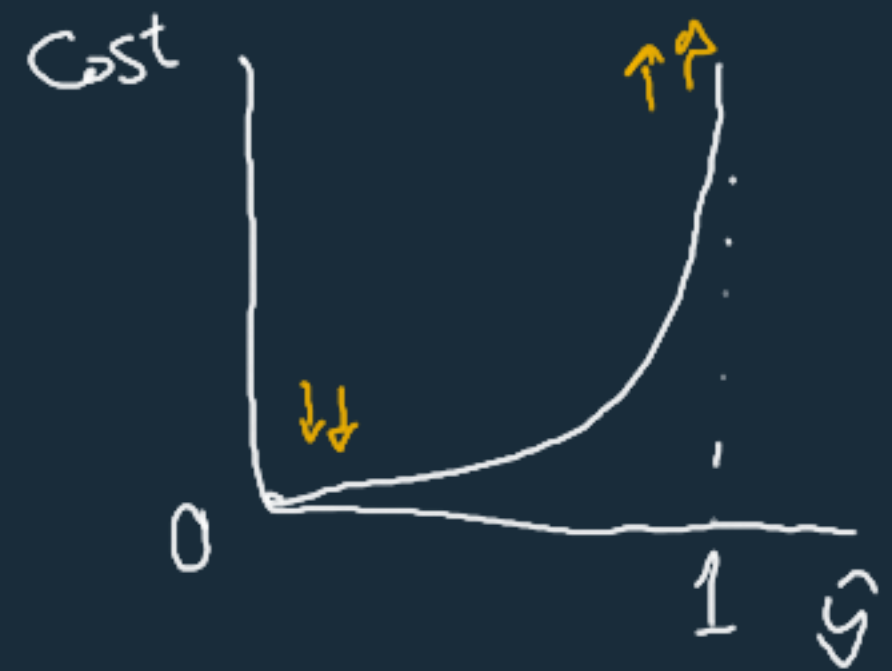
Cost fn

Log. Regression

$$J = -y \log(\hat{y}) - (1-y) \log(1-\hat{y})$$



$$-\log(1-\hat{y}) \quad y=0$$



$$\bar{J} = \frac{1}{m} \sum_{i=1}^m y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) (1 - \log(\hat{y}^{(i)}))$$

Cost fn
for
Log. Regression

$$\bar{J}(\theta) = \frac{1}{m} \sum_{i=1}^m y^{(i)} \log(\sigma(\theta^T x^{(i)})) + (1 - y^{(i)}) (1 - \log(\sigma(\theta^T x^{(i)})))$$



Min. $\rightarrow \theta$ (Best parameters)

XX Normal Equation XX

Complex



Gradient
Descent

Random initialization for θ

$\bar{J}(\theta)$

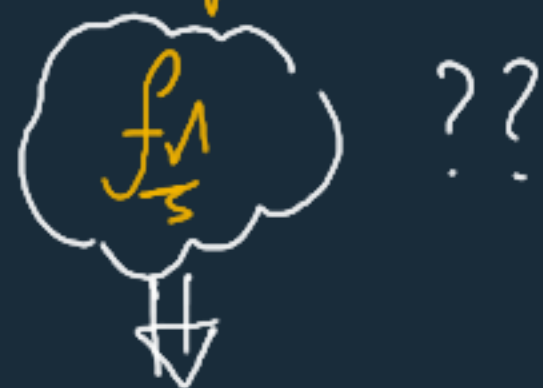
$$\theta^{\text{new}} = \theta^{\text{old}} - \alpha \frac{\partial \bar{J}}{\partial \theta}$$

iter.

$$\theta \rightsquigarrow \sigma(\theta^T x)$$

Log. Reg. Model

Multi-Class Classification



$[p(0) \quad p(1) \quad p(2)]$

class 1 ←

~ 0 ←

~ 2 ←

0.2

0.7

0.1

0.7

0.2

0.1

0.2

0.1

0.7

Verginica

False

True

$[p(0) \quad p(1)]$

0.7

0.3

$[p(0) \quad p(1)]$

0.7

0.3

Binary
clf.



Sigmoid → (0 → 1)



$P(y=1)$

Handwritten Digits
0 → 9
10 classes

Multi-class Strategies

① one-vs-all / one-vs-rest "ovr"

↳ # Binary clfs = # classes



Iris → 3 Binary clfs

Setosa vs. Not Setosa



Virginica vs Not Virginica



Versicolor vs. Not Versicolor



② one-vs-one "ovo"

$0 \rightarrow 1$
 $0 \rightarrow 2$
 $1 \rightarrow 2$

} 3 clf



"OVR" $\rightarrow$ 5 Binary clfs

5 classes [0, 1, 2, 3, 4]

"OVO" $\rightarrow$ 10 Binary clfs

$0 \rightarrow 1$
 $0 \rightarrow 2$
 $0 \rightarrow 3$
 $0 \rightarrow 4$
 $1 \rightarrow 2$
 $1 \rightarrow 3$
 $1 \rightarrow 4$
 $2 \rightarrow 3$
 $2 \rightarrow 4$
 $3 \rightarrow 4$

OVO

$$\# \text{ clfs} = \frac{n(n-1)}{2}$$

$n \rightarrow \# \text{ classes}$

③ Softmax

$$p(0) \quad p(1) \quad p(2) \quad \dots \quad p(n)$$

Binary
(Sigmoid)

$$Z = \theta^T X$$

$$p = \sigma(Z)$$

$$\rightarrow \text{Cont. } 1.65$$

$$\quad \quad \quad \downarrow$$

$$\quad \quad \quad 0.5$$

Softmax

Score

$$\left. \begin{array}{l} s_1 = \theta_1^T X \\ s_2 = \theta_2^T X \\ \vdots \\ s_n = \theta_n^T X \end{array} \right\} \begin{array}{l} \text{Cont.} \\ \checkmark \\ \checkmark \end{array}$$

$$\frac{e^{s_i}}{\sum_{i=1}^n e^{s_i}}$$

$$\left\{ \begin{array}{l} p(1) \\ p(2) \\ \vdots \\ p(n) \end{array} \right.$$

Linear Regression



overfitting



Regularization

Control

$\alpha \propto \text{Reg.}$

↑

↓

↑

↓

Regularization



Underfitting

↑

Train Error

↑

Test Error



Good fit

✓

✓



overfitting

↓

↓

poor Generalization

Penalty $\begin{cases} \rightarrow L1 \\ \rightarrow L2 \end{cases}$

Control $\rightarrow C \propto \frac{1}{\text{Regulariz.}}$

↑

↓